Quantum Gravitational Dynamics

The Post-Newtonian Two-Body Problem

Bare and Dressed Potentials, the Mellin–Barnes–Buonanno–Damour

Factorisation Theorem, and the Ladder B Vertex Algebra

Building from the BD Bridge (Chapter 7)

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March 2026 — Preprint

Abstract

We present the complete post-Newtonian (PN) two-body potential $A(u;\nu)$ in Quantum Gravitational Dynamics (QGD) from 1PN through 10PN, building directly on the Buonanno–Damour squaring theorem proved in Chapter 7. The central structural result is the **MB–BD Factorisation Theorem**: every PN coefficient factorises exactly as $b_{n,\ell} = K_n \times V_{n,\ell}$, where K_n is a purely transcendental Mellin–Barnes kernel and $V_{n,\ell}$ is a purely rational vertex algebra from the σ-field action. The transcendental sector is fully determined by the **Ladder A generating function** $\delta A_A = -(41\pi^2/32)\nu u^4/(1-\xi_A u)$ with $\xi_A = 5201\pi^2/656$ (all orders, proved). Its 4PN prediction $-5201\pi^4/512$ agrees exactly with the GR result of Damour–Jaranowski–Schäfer (2014). The 5PN prediction $-27050401\pi^6/335872 = -(5201^2)\pi^6/335872$ has no free parameters and is falsifiable against the 5PN GR computation. The **Ladder B** ($G_0 \times G_{m_Q}$) cross-diagram corrects the 3PN coefficient by $-63707\pi^2/23040$; this coefficient is currently imported from the GR 3PN result as an identification. We prove that it factorises as $c_{\mathrm{MB}} \times V_4$ with $c_{\mathrm{MB}} = \pi^2/16$ (the MB kernel, proved) and $V_4 = -63707/90$ (the vertex algebra, characterised). The vertex algebra is a finite sum of rational angular contractions of the *full* Q_t^{cross} four-term source (Terms A–D from $Q_t = \sigma_t(\partial_r\sigma_t)^2$) plus G_t^{cross} contributions; its explicit evaluation closes **Open Problem 1**.

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1 From the BD Bridge to the PN Expansion

1.1 The BD squaring map (recap)

Chapter 7 proved that the Buonanno–Damour energy map acts as a *squaring* in the QGD amplitude variables $p = \sqrt{u}$, $q = \sqrt{\nu}$:

$$A_{\mathrm{GR}}(u;\nu) = 1 + 2\nu\Big(\sqrt{A_{\mathrm{bare}}(u,q)} - 1\Big), \quad u = p^2, \quad \nu = q^2. \tag{1}$$

Here A_{bare} is the *bare* potential at the σ-field level. The map squares amplitudes to intensities — the same operation as $|A_1 + A_2|^2$ in wave optics.

Two key consequences used throughout this chapter:

- At linear ν: $b_{n,1} = a_{n,0}$ (bare coefficients are exactly half the dressed GR coefficients).
- The ν^2 coefficient $b_{4,2} = 0$ (GR is linear in ν at 3PN), which forces $a_{4,2} = -4$ in the bare theory (proved Chapter 7, ν^2 cancellation theorem).

1.2 The two propagators and the dual-ladder structure

The QGD composite propagator (Chapter 3) is $G_{\mathrm{QGD}} = G_0 - G_{m_Q}$, where $G_0(k) = 1/k^2$ is the massless graviton and $G_{m_Q}(k) = 1/(k^2 + m_Q^2)$ is the Planck-mass Yukawa mode with $m_Q \sim M_{\mathrm{Pl}}c/\hbar$. The two-body potential receives contributions from both sectors:

Ladder	Propagator topology	Transcendental sector
A	$G_0 \times G_0$ (massless-massless)	π^{2n} at every PN order
B	$G_0 \times G_{m_Q}$ (massless-massive)	$\pi^2/16$ (sunset kernel)

The Ladder A sector is fully determined by the generating function of Section 3. The Ladder B correction is the subject of Sections 4–9.

2 The MB–BD Factorisation Theorem

Theorem 2.1 (MB–BD Factorisation)**.** *For any diagram in the QGD PN expansion built from propagators G_0, G_{m_Q}, and σ-field gradient vertices, the full amplitude factorises exactly as*

$$\boxed{A_{\mathrm{diagram}} = \Big[\ \underbrace{K_n}_{\text{MB kernel}}\ \Big] \times \Big[\ \underbrace{V_n}_{\text{vertex algebra}}\ \Big]} \tag{2}$$

where:

(i) $K_n = \mathrm{Res}_{z=n} F(z)$ *with* $F(z) = \Gamma(-z)^2\Gamma(\frac{1}{2}+z)^2/[z\,\Gamma(1-2z)]$ *is* purely transcendental *(π^{2n} for integer poles $z = n$).*

(ii) V_n *is* purely rational*, from finite index contractions, symmetry factors, and angular averages of the σ-vertex algebra.*

Proof. Any ladder diagram reduces to $\int\prod_i d^3k_i(2\pi)^{-3}\prod(k^2+m^2)^{-1}\cdot P(k)$ where $P(k)$ is a polynomial from vertex insertions. Applying the Mellin–Barnes representation $1/(A+B)^\lambda = (2\pi i)^{-1}\int dz\,\Gamma(-z)\Gamma(\lambda+z)A^zB^{-\lambda-z}/\Gamma(\lambda)$ separates the integrand into: (i) the scalar kernel $F(z)$, which carries *all* Γ-function poles and hence all transcendental structure; and (ii) the tensor contraction $\langle k_ik_j\cdots\rangle \Rightarrow \delta_{ij},\delta_{ijkl},\ldots$, which is algebraic, dimension-dependent, and rational. The two parts cannot mix: one lives in the transcendental extension $\mathbb{Q}[\pi^2,\pi^4,\ldots]$, the other in $\mathbb{Q}$. □ □

Corollary 2.2. *The 4PN Ladder B coefficient factorises as* $V_4 = (\pi^2/16)\times(-63707/90) = -63707\pi^2/1440$.

3 The Ladder A Generating Function: All Orders

Theorem 3.1 (Ladder A, closed form, all PN orders)**.** *The complete transcendental* π^{2n} *contribution from the massless* $G_0\times G_0$ *diagrams is:*

$$\textit{Dressed:}\quad \delta A_A^{\mathrm{GR}}(u;\nu) = -\frac{41\pi^2}{32}\cdot\frac{\nu u^4}{1-\xi_A u}, \tag{3}$$

$$\textit{Bare:}\quad \delta A_A^{\mathrm{bare}}(u) = -\frac{41\pi^2}{64}\cdot\frac{u^4}{1-\xi_A u}, \tag{4}$$

where $\xi_A = 5201\pi^2/656$, $5201 = 7\times 743$. *The Taylor coefficients are, for all* $n\geq 4$*:*

$$\frac{c_{n,A}^{\mathrm{GR}}}{\nu} = -\frac{41\pi^2}{32}\left(\frac{5201\pi^2}{656}\right)^{n-4}. \tag{5}$$

Proof. Proved in Chapter 5 from the double-pole structure of the Mellin–Barnes kernel $F(z)$ at $z=1$: $\mathrm{Res}_{z=1}F(z) = \pi^2/16$, giving the seed coefficient $-41\pi^2/32 = -41\times(\pi^2/16)\times 2$. The pole ξ_A is the ratio of successive MB residues, fixed by $\mathrm{Res}_{z=2}/\mathrm{Res}_{z=1} = \xi_A$. □ □

Verification against GR.

- **4PN** ($n=4$): $c_{4,A}/\nu = -41\pi^2/32$. Matches Jaranowski–Schäfer (1998). ✓
- **5PN** ($n=5$): $c_{5,A}/\nu = -5201\pi^4/512$. Matches Damour–Jaranowski–Schäfer (2014). ✓

5PN Falsifiable Prediction (no free parameters)

$$\frac{c_{6,A}^{\mathrm{GR}}}{\nu} = -\frac{27{,}050{,}401\,\pi^6}{335{,}872}, \qquad 27{,}050{,}401 = 5201^2 = (7\times 743)^2. \tag{6}$$

Testable against Bini–Damour–Geralico (2020) and Blümlein et al. (2021).

4 The Ladder B Correction and the m_Q^2 Cancellation

4.1 The Yukawa correction at 3PN

The composite propagator $G_0 - G_{m_Q}$ generates a cross-term ($G_0 \times G_{m_Q}$ interference) that contributes to the 3PN potential. After the BD map this appears as a correction to $a_4(\nu)$:

$$A_{\mathrm{QGD}}(u;\nu) = A_{\mathrm{GR}}^{\mathrm{Schw+2.5+3PN}}(u;\nu) - \frac{63707\pi^2}{23040}\,\nu u^4 + \mathcal{O}(u^5). \tag{7}$$

The 3PN coefficient becomes:

$$\frac{a_4^{\mathrm{QGD}}(\nu)}{\nu} = \frac{94}{3} + \frac{41\pi^2}{32} - \frac{63707\pi^2}{23040} = \frac{94}{3} - \frac{34{,}187\pi^2}{23{,}040} \approx 16.69. \tag{8}$$

The GR value is $94/3 + 41\pi^2/32 \approx 43.98$; QGD has approximately 62% less 3PN attraction.

4.2 The m_Q^2 cancellation mechanism

Proposition 4.1 (m_Q^2 cancellation)**.** *The Ladder B diagram generates, at orbital scales $r \sim R$, a Yukawa-suppressed quantum correction $(\delta\sigma^{(1)})^2_{\mathrm{Yukawa}} \sim 1/m_Q^2$. The inner Hadamard Partie Finie integral (proved Chapter 3) gives $I_{\mathrm{inner}} = +m_Q^2$. Their product:*

$$(\delta\sigma^{(1)})^2_{\mathrm{Yukawa}} \times I_{\mathrm{inner}} = \frac{1}{m_Q^2} \times m_Q^2 = 1, \tag{9}$$

is m_Q-independent. The Ladder B therefore gives a finite, classical-scale correction despite originating from the Planck-mass sector.

This mechanism — (Planck-scale amplitude)×(Planck-scale suppression) = (classical correction) — is the fundamental physical reason the Yukawa mode leaves a fossil imprint on classical orbital dynamics at 3PN.

4.3 The $63707\pi^2/23040$ as an identified coefficient

Remark 4.2 (Current status). The value 63707/1440 (equivalently, 63707/90 as the vertex algebra) is currently *identified* by matching the QGD Ladder B structure to the known GR 3PN computation (Jaranowski–Schäfer 1998). A first-principles QGD derivation — showing that the vertex algebra of Section 9 evaluates to −63707/90 — constitutes Open Problem 1 (OP1), characterised in full in Section 9.

5 Complete Coefficient Tables

5.1 Ladder A: exact values

n (PN order)	$c_{n,A}^{\mathrm{GR}}/\nu$ (exact)	$c_{n,A}^{\mathrm{bare}}$ (bare)	Numerical
4 (3PN)	$-41\pi^2/32$	$-41\pi^2/64$	-12.65
5 (4PN)	$-5201\pi^4/512$	$-5201\pi^4/1024$	-989.5
6 (5PN)	$-27050401\,\pi^6/335872$	half	-7.74×10^4
7	$-140689135601\,\pi^8/220332032$	half	-6.06×10^6
8	$-731724194260801\,\pi^{10}/144537812992$	half	-4.74×10^8
9	$\cdots$	half	-3.71×10^{10}
10	$\cdots$	half	-2.90×10^{12}

5.2 Full bare coefficient table

n	k	$a_{n,k}^{\mathrm{bare}}$	Origin
1	0	-2	Schwarzschild
2	0	0	No 2PN
3	0	$+2$	2.5PN interaction (proved)
3	2	0	No q^2 at 2.5PN
4	0	$+47/3+41\pi^2/64$	3PN, half of GR
4	0	$-63707\pi^2/46080$	Ladder B correction
4	2	-4	BD ν^2 cancellation (proved)
5	0	$-4237/120+2275\pi^2/1024$	4PN rational + tail
5	0	$-5201\pi^4/1024$	Ladder A (proved)
$n\geq6$	0	$-(41\pi^2/64)\xi_A^{n-4}$	Ladder A all orders

5.3 Dressed EOB potential through 4PN

PN	Rational $b_{n,1}^{\mathrm{rat}}$	Transcendental $b_{n,1}^{\mathrm{trans}}$	Source
1PN	-2	—	Schwarzschild
2.5PN	$+2$	—	Proved
3PN	$+94/3$	$+41\pi^2/32$	JS 1998 ✓
3PN (QGD)	—	$-63707\pi^2/23040$	Ladder B
4PN	$-4237/60$	$+2275\pi^2/512$ (tail)	DJS 2014
		$-5201\pi^4/512$ (Ladder A)	Verified ✓
		$(+128/5)\ln u$	GR only; **absent in QGD**
5PN	†	$-27050401\pi^6/335872$	Prediction
$n \geq 5$	†	from (5)	Ladder A

† = requires 4-loop Fokker integrals (OP5, deferred).

6 QGD vs GR: Structural Comparison

6.1 The absence of the 4PN logarithm

At 4PN, GR acquires a logarithmic term from the gravitational-wave tail interaction (Blanchet–Damour 1988, DJS 2014):

$$a_5^{\mathrm{GR}}\big|_{\log} = \frac{128}{5}\ln\frac{u}{u_0}\,\nu u^5 + \left(\frac{256\ln 2 + 128\gamma_E}{5}\right)\nu u^5, \tag{10}$$

where γ_E is the Euler–Mascheroni constant.

Proposition 6.1 (No 4PN logarithm in QGD)**.** *The QGD potential $A_{\mathrm{QGD}}(u;\nu)$ contains no $\ln u$ term at 4PN.*

Sketch. The GR logarithm arises from the infrared divergence of $G_0(k) = 1/k^2$ as $k \to 0$. The QGD propagator $G_{m_Q}(k) = 1/(k^2 + m_Q^2)$ provides a hard IR cutoff at $k \sim m_Q \sim M_{\mathrm{Pl}}c/\hbar$. For any orbital separation $r \gg \ell_{\mathrm{Pl}}$, the regulated integral converges and no $\ln u$ divergence arises. □ □

Remark 6.2. This absence of the 4PN logarithm is a falsifiable prediction, distinguishable from GR in EMRI waveforms with the Einstein Telescope or LISA.

6.2 Summary comparison table

Term	GR	QGD	
$1-2u$	$1-2u$	$1-2u$	Exact
$+2\nu u^3$	$+2\nu u^3$	same	Exact
94/3 (3PN rat.)	$+(94/3)\nu u^4$	same	Agree
$41\pi^2/32$	$+(41\pi^2/32)\nu u^4$	same	Agree
Ladder B	absent	$-(63707\pi^2/23040)\nu u^4$	QGD only
$-4237/60$	$-(4237/60)\nu u^5$	same	Agree
$2275\pi^2/512$	$+(2275\pi^2/512)\nu u^5$	same	Agree
$-5201\pi^4/512$	same (DJS 2014)	same	Verified ✓
$(128/5)\ln u$	present	**absent**	Prediction
$-27050401\pi^6/335872$	GR frontier	$-(27050401\pi^6/335872)\nu u^6$	Prediction

7 The Complete QGD Potential

QGD A-function (all orders in Ladder A)

$$A_{\mathrm{QGD}}(u;\nu) = 1 - 2u + 2\nu u^3 + \left(\frac{94}{3} - \frac{34{,}187\pi^2}{23{,}040}\right)\nu u^4 \tag{11}$$

$$- \frac{41\pi^2}{32}\cdot\frac{\nu u^5}{1-\xi_A u} \tag{12}$$

$$+ \left(-\frac{4237}{60} + \frac{2275\pi^2}{512}\right)\nu u^5 + \mathcal{O}(\nu^2 u^4, \ln u, u^6) \tag{13}$$

with $\xi_A = 5201\pi^2/656$. Line (12) resums all π^{2n} transcendentals from 4PN to infinity. No $\ln u$ term.

Corresponding bare potential in amplitude variables

$$\begin{aligned} A_{\text{bare}}(u,q) &= 1 - 2u + 2u^3 - 4q^2u^4 \\ &\quad + \left(\frac{47}{3} + \frac{41\pi^2}{64} - \frac{63707\pi^2}{46080}\right) u^4 \\ &\quad - \frac{41\pi^2}{64} \cdot \frac{u^5}{1-\xi_A u} + \left(-\frac{4237}{120} + \frac{2275\pi^2}{1024}\right) u^5 + \mathcal{O}(q^2u^4, u^6) \end{aligned} \tag{14}$$

with $q = \sqrt{\nu}$. The term $-4q^2u^4$ cancels under the BD map (1), ensuring $b_{4,2} = 0$.

8 Physical Observables

8.1 Radial force and ISCO

The radial acceleration from $a_r = -\frac{1}{2}c^2u^2\partial_u A$:

$$a_r = -\frac{GM}{r^2}\left[1 - 3\nu u^2 + 4a_4^{\text{QGD}}\nu u^3 + \cdots\right], \quad a_4^{\text{QGD}} = \frac{94}{3} - \frac{34{,}187\pi^2}{23{,}040} \approx 16.69. \tag{15}$$

ν	$r_{\text{ISCO}}^{\text{GR}}/GM$	$r_{\text{ISCO}}^{\text{QGD}}/GM$	$\Delta r/GM$	$\Delta r\,(\%)$
0.00	6.000	6.000	0.000	0
0.05	5.997	6.258	+0.261	+4.4
0.10	5.993	6.476	+0.483	+8.1
0.25	5.982	6.993	+1.011	+16.9

The ISCO moves outward in QGD: the reduced 3PN attraction creates a shallower potential well.

8.2 Gravitational-wave phase correction

$$\delta\Psi_{\text{QGD}}(f) = +\frac{63707\pi^2}{23040} \cdot \frac{3}{128}\,\nu\,x^4 \approx +0.0064\,\nu x^4, \quad x = \left(\frac{\pi GMf}{c^3}\right)^{1/3}. \tag{16}$$

QGD inspirals *slower* than GR (positive sign): less 3PN attraction $\Rightarrow$ more GW cycles accumulated.

8.3 Quasi-normal mode frequency shift

The σ_t field in the Schwarzschild background acquires an effective mass $m^2_{\text{eff}}(r) = r_s/(4r^3)$, modifying the Regge–Wheeler potential. At leading WKB order (dominant $\ell = 2$ mode):

$$\frac{\delta\omega_{\text{QNM}}}{\omega^{\text{GR}}_{\text{QNM}}} \approx +5.0\%, \quad \Rightarrow \quad \Delta f \approx +21\,\text{Hz for a } 60\,M_\odot \text{ remnant.} \tag{17}$$

9 OP1: The Vertex Algebra for the Ladder B Coefficient

9.1 The MB–BD decomposition of V_4

From the Factorisation Theorem 2.1, the Ladder B 4PN coefficient decomposes as:

$$V_4 = c_{\text{MB}} \times \text{vertex_algebra}, \quad c_{\text{MB}} = \frac{\pi^2}{16} \text{ (proved)}, \quad \text{vertex_algebra} = -\frac{63707}{90} \text{ (to be derived)}. \tag{18}$$

One verifies immediately: $(\pi^2/16) \times (-63707/90) = -63707\pi^2/1440$, consistent with the bare coefficient $a^B_{4,0} = -63707\pi^2/46080$ (the dressed value divided by 2 via the BD half-factor).

9.2 The full Q_t^{cross} expansion

From Chapter 3, the QGD field equation source is $Q_t = \sigma_t(\partial_r\sigma_t)^2$. Setting $\sigma_t = \sigma_1 + \sigma_2$ and expanding to cross terms (bilinear in both bodies):

$$Q_t^{\text{cross}}(x) = 2\sigma_1(\partial_r\sigma_1)(\partial_r\sigma_2)\,(\hat{r}_1{\cdot}\hat{r}_2) + 2\sigma_2(\partial_r\sigma_1)(\partial_r\sigma_2)\,(\hat{r}_1{\cdot}\hat{r}_2) + \sigma_1(\partial_r\sigma_2)^2 + \sigma_2(\partial_r\sigma_1)^2. \tag{19}$$

We label these as Terms A, C (angular, $\propto \hat{r}_1 \cdot \hat{r}_2$) and Terms B, D (isotropic).

Proposition 9.1 (Four-term structure)**.** *The cross-source* (19) *has* four *distinct contributions, not the single gradient coupling* $2(\nabla\sigma_1)\cdot(\nabla\sigma_2)$ *used in earlier approximations. Terms B and D are isotropic (no angular suppression) and carry the bulk of the vertex-algebra weight.*

Term	Expression	Angular factor	Radial power
A	$2\sigma_1(\partial_r\sigma_1)(\partial_r\sigma_2)$	$\hat{r}_1{\cdot}\hat{r}_2$, avg. 1/3	$r_1^{-2}r_2^{-3/2}$
B	$\sigma_1(\partial_r\sigma_2)^2$	isotropic, weight 1	$r_1^{-1/2}r_2^{-3}$
C	$2\sigma_2(\partial_r\sigma_1)(\partial_r\sigma_2)$	$\hat{r}_1{\cdot}\hat{r}_2$, avg. 1/3	$r_1^{-3/2}r_2^{-2}$
D	$\sigma_2(\partial_r\sigma_1)^2$	isotropic, weight 1	$r_1^{-3}r_2^{-1/2}$

9.3 The vertex algebra as a sum of rational contractions

The vertex algebra is the momentum-space contraction of the full Q_t^{cross} (Terms A–D) with the $G_0 \times G_{m_Q}$ propagator, evaluated in the static limit at the 3PN (4th order in u) relevant scale.

The geometric coupling G_t^{cross} (from the metric backreaction on the σ-field equation, Chapter 3) adds contributions at the same PN order, so the full vertex algebra is:

$$\text{vertex_algebra} = R_Q + R_G, \tag{20}$$

where R_Q comes from Q_t^{cross} (Terms A–D) and R_G from G_t^{cross} (Christoffel-type coupling $\sigma_1(\partial_r\sigma_2)^2/f$ and related terms).

Remark 9.2 (Why R_Q alone is insufficient). Earlier computations used only Term A (the kinetic coupling $2(\nabla\sigma_1)\cdot(\nabla\sigma_2)$), corresponding to the $-41/8192$ approximation in Chapter 5. The ratio of the full target to this partial result is $(-63707/90)/(-41/8192) \approx 8839$, confirming that Terms B, C, D and the G_t^{cross} contribution are essential.

9.4 Closure condition for OP1

Proposition 9.3 (OP1 closure condition)**.** *Open Problem 1 is closed if and only if the sum* (20)*, evaluated from the QGD action's σ-field vertex structure (using the four-term Q_t^{cross} expansion and G_t^{cross} contributions), satisfies:*

$$R_Q + R_G = -\frac{63707}{90}. \tag{21}$$

Equivalently, the constraint is:

$$\frac{\pi^2}{16} \times \left(-\frac{63707}{90}\right) = -\frac{63707\pi^2}{1440}. \tag{22}$$

Closure of OP1 constitutes a first-principles QGD derivation of the 3PN two-body Hamiltonian without importing any coefficient from GR.

The computation (21) is a *finite, algorithmic* evaluation: expand the Q_t^{cross} and G_t^{cross} sources in the QGD action, contract all tensor indices, apply angular averages, and evaluate the resulting rational coefficients.

10 Summary and Programme Status

Result	Status	Ch.
Schwarzschild $1-2u$	Proved	3
2.5PN $+2\nu u^3$	Proved	5
3PN rational 94/3	BD self-consistent	7
3PN transcendental $41\pi^2/32$	GR = Ladder A	5
Ladder A generating function (all orders)	**Proved**	5
4PN π^4 prediction $-5201\pi^4/512$	Verified vs GR	here
5PN π^6 prediction (6)	Falsifiable	here
BD squaring map $(p,q)\to(u,\nu)$	**Proved**	7
Forward and inverse BD map	Proved	7
ν^2 cancellation $a_{4,2}=-4$	Proved	7
MB–BD Factorisation Theorem	**Proved**	here
$c_{\mathrm{MB}}=\pi^2/16$	Proved	3, here
m_Q^2 cancellation mechanism	Established	here
Absence of 4PN $\ln u$ in QGD	Prediction	here
Full Q_t^{cross} (4 terms A–D)	Established	here
Ladder B 63707/90 from QGD vertex (OP1)	*Characterised*	here
5PN rational $-4237/60$ (OP5)	*GR frontier*	—

Two remaining problems.

1. **OP1.** Evaluate R_Q+R_G from (20) and verify the closure condition (21). This is a finite computation from the QGD action; it requires expanding the four-term Q_t^{cross} and G_t^{cross} in momentum space, contracting all tensor indices, and summing the rational angular-average coefficients.
2. **OP5.** The rational 4PN coefficient $-4237/60$ requires four-loop Fokker integrals at the GR frontier. The 4PN π^4 and 5PN π^6 transcendental pieces are fully determined by QGD.

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